

PAPER_483 — Multi-System UQFF Compiler Modules: 4/5/7/8 AstroSystems Architecture

Abstract

This paper documents the **multi-system compiler modules** extracted from `grok_share_bdfb3a05b06.txt` — a new architectural tier above individual-system modules. Where PAPER_481 documents single-system UQFF implementations, these modules simultaneously compute $F_{U_{Bi_i}}$ for 4, 5, 7, and 8 astrophysical systems through a single `computeMasterEquations(system)` call, using a runtime-dispatched parameter lookup (`setSystemParams`). New physics includes the triadic simultaneous solver (`computeTriadicSolution`), gas-nebula integration (`computeGasNebulaIntegration`), and the Source160 computation validation showing $F_{U_{Bi_i}} \approx -8.32 \times 10^{217}$ N universally for template-mass systems.

Source: `grok_share_bdfb3a05b06.txt`, docx: “4 Astro Systems_11Oct2025”, “5 Astro Systems_11Oct2025”, “7 Astro Systems_11Oct2025”, “8 Astro Systems_11Oct2025”, “8 Astro Systems_B_11Oct2025”, “Source160.docx”. Analyzed Oct 23, 2025.

1. Module Hierarchy

UQFF Multi-System Architecture

AstroSystemsUQFFModule (4 systems)

Systems: NGC685, NGC3507, NGC3511, AT2024tvd

New API: `computeMasterEquations(system)`, `setSystemParams(system)`

`computeGravityCompressed(system)`, `computeResonanceUr(U_dp, U_r, system)`

`computeBuoyancyUbi(system)`

UQFFNebulaTriadicModule (5-7 systems, 2 canonical versions)

Systems v5: NGC3596, NGC1961, NGC5335, NGC2014 + Carina

Systems v7: NGC685, NGC3507, NGC3511, Carina + 3 extended

New API: `computeGasNebulaIntegration(system)` ← unique nebular term

UQFFBuoyancyModule 7-system (multi-system buoyancy variant)

Systems: M74, M16, M84, Centaurus A (4+ systems)

Source: “7 Astro Systems_11Oct2025”

UQFF8AstroSystemsModule (8 systems)

Systems: NGC4826, NGC1805, NGC6307, NGC7027 + 4 more

New API: full 8-system dispatch via `computeMasterEquations(system)`

UQFF8AstroTriadicModule (8 systems, triadic solver)

Systems: Same 8 as above

New API: `computeTriadicSolution(system, t)` ← simultaneous triadic solve

`computeGravityCompressed(system)`, `computeResonanceUr(U_dp, U_r, system)`

`computeBuoyancyUbi(system)`

2. System-Specific Parameter Registries

2.1 AstroSystemsUQFFModule (4 Systems)

Comment doc: "Multi-system params: NGC685 M=1e41 kg r=1e21 m; NGC3507 M=2e41 kg r=2e21 m; NGC3511 M=3e41 kg r=3e21 m; AT2024tvd M=1..."

System	M (kg)	r (m)	Type
NGC685	10^{41}	10^{21}	Barred spiral galaxy
NGC3507	2×10^{41}	2×10^{21}	Barred lenticular galaxy
NGC3511	3×10^{41}	3×10^{21}	Spiral galaxy
AT2024tvd	10^{41} (template)	10^{21}	Sept 2025 transient TDE off-nucleus

AT2024tvd is notable — this is the off-nucleus tidal disruption event discovered September 2024 in a galaxy 600 Mpc away, consistent with UQFF cluster-scale parameters.

2.2 UQFFNebulaTriadicModule (5 Systems)

Comment doc: "Multi-system params: NGC3596 M=1e41 kg r=1e21 m; NGC1961 M=2e41 kg r=2e21 m; NGC5335 M=3e41 kg r=3e21 m; NGC2014 M=4e..."

System	M (kg)	r (m)	Type
NGC3596	10^{41}	10^{21}	Spiral galaxy (Leo cluster)
NGC1961	2×10^{41}	2×10^{21}	Peculiar spiral (AGN)
NGC5335	3×10^{41}	3×10^{21}	Lenticular galaxy
NGC2014	4×10^{41}	$\sim 4 \times 10^{21}$	LMC H II region
Carina	$\sim 10^{36}$	$\sim 10^{17}$	Carina Nebula star-forming region

2.3 UQFFNebulaTriadicModule (7 Systems — Canonical Version)

Comment doc: "Multi-system params: NGC685 M=1e41 kg r=1e21 m; NGC3507 M=2e41 kg r=2e21 m; NGC3511 M=3e41 kg r=3e21 m; Carina M=1e36..." - 4 systems from 4AstroSystems + Carina + 2 additional (from 7-system batch)

2.4 UQFF8AstroSystemsModule (8 Systems)

Comment doc: "Multi-system params: NGC4826 M=1e41 kg r=1e21 m; NGC1805 M=2e41 kg r=2e21 m; NGC6307 M=3e41 kg r=3e21 m; NGC7027 M=..."

System	M (kg)	r (m)	Type
NGC4826	10^{41}	10^{21}	Black Eye Galaxy
NGC1805	2×10^{41}	2×10^{21}	Open cluster in LMC
NGC6307	3×10^{41}	3×10^{21}	Elliptical galaxy
NGC7027	—	—	Planetary nebula (JWST 2022)
+4 more	—	—	From “8 Astro Systems_B” batch

3. New API Methods

3.1 computeMasterEquations(system) — Multi-System Dispatcher

```
cdouble AstroSystemsUQFFModule::computeMasterEquations(const std::string& system) {
    setSystemParams(system); // Populate variables["M"], variables["r"] for chosen system
    return computeF(t_default); // Full F_U_Bi_i for that system's parameters
}
```

Supports runtime system switching:

```
AstroSystemsUQFFModule mod;
auto F_NGC685 = mod.computeMasterEquations("NGC685");
auto F_NGC3507 = mod.computeMasterEquations("NGC3507");
auto F_AT2024 = mod.computeMasterEquations("AT2024tvd");
```

3.2 computeGasNebulaIntegration(system) — NEW (UQFFNebulaTriadicModule only)

Unique to the nebular triadic module — computes additional gas-phase integration term:

$$F_{gas} = \int \rho_{gas}(r) \cdot g_{UQFF}(r, t) dV \approx \rho_{gas} \cdot g_{UQFF} \cdot V_{nebula}$$

This term accounts for the ISM/gas column density contribution to the UQFF total field in nebular environments, scaling with $L_{H\alpha}$ and SFR. Specific to NGC3596, NGC1961, NGC5335, NGC2014, and Carina Nebula — all star-forming or nebular gas systems.

3.3 computeResonanceUr(U_dp, U_r, system) — Resonance Mode Selector

```
cdouble AstroSystemsUQFFModule::computeResonanceUr(int U_dp, int U_r, const std::string& system) {
    // U_dp = DPM resonance mode (0=off, 1=on)
    // U_r = resonant coupling mode (0=magnetic, 1=vacuum)
    // system = dispatch key
}
```

Extends DPM resonance with two independent control flags for fitting observational constraints.

3.4 computeTriadicSolution(system, t) — NEW (UQFF8AstroTriadicModule only)

Simultaneously solves all three triadic arms for a given system and time:

$$\vec{F}_{triadic} = (F_{\text{compressed}}(t), F_{\text{resonant}}(t), F_{\text{buoyancy}}(t))$$

Returns the triadic solution vector as three `cdouble` values, enabling simultaneous constraint by compressed, resonant, and buoyancy observational data.

4. Source160 Validation Data (Thoughts L10436)

From the Grok Thoughts section analyzing Source160.docx:

4.1 Full $F_{U_{Bi_i}}$ (template-mass systems)

$$F_{U_{Bi_i}} \approx -8.32 \times 10^{217} + i(-6.75 \times 10^{160}) \text{ N}$$

Identical for: **Tycho's SNR**, **Abell 2256**, **Tarantula Nebula**, **NGC 253** — when evaluated at template-mass ($M = 10^{41} \text{ kg}$, $r = 10^{21} \text{ m}$, $\omega_0 = 10^{-15}$).

4.2 Compressed Integrand per System

$$F_{U_{Bi_i}, \text{integrand}} = \sum \text{terms} \approx \begin{cases} 6.16 \times 10^{45} \text{ N} & \text{Abell 2256 } (M = 1.23 \times 10^{45} \text{ kg}) \\ 6.16 \times 10^{39} \text{ N} & \text{Template mass systems} \end{cases}$$

4.3 DPM Resonance per System

$$\text{DPM}_{res} = \frac{g_L \mu_B B_0}{\hbar \omega_0} \approx \begin{cases} 1.76 \times 10^{15} & \text{Tycho's SNR} \\ 1.76 \times 10^{17} & \text{Abell 2256} \\ 1.76 \times 10^{18} & \text{Tarantula, NGC 253} \end{cases}$$

4.4 Buoyancy U_{b1} (all systems universal)

$$U_{b1} = \beta_i V_{infl,UA} \rho_{vac,A} a_{univ} \approx (6 \times 10^{-19} + 6.6 \times 10^{-20}i) \text{ N}$$

4.5 Superconductive U_i (all systems, $t = t_0$)

$$U_i \approx (1.7 \times 10^{-43} + 1.2 \times 10^{-44}i) \text{ N}$$

5. Architecture Notes

5.1 File Inventory (Created Session 126)

File	Size (chars)	Notes
AstroSystemsUQFFModule3587	1,587	4-system dispatcher
AstroSystemsUQFFModule4941p	4,941	w/ setSystemParams, computeGravity-Compressed

File	Size (chars)	Notes
UQFFNebulaTriadicModule.h	3170h	5+7-system, computeGasNebulaIntegration
UQFFNebulaTriadicModule.cpp	11343cpp	canonical largest version
UQFF8AstroSystemsModule.h	4114h	8-system dispatcher
UQFF8AstroSystemsModule.cpp	11212cpp	largest 8-system implementation
UQFF8AstroTriadicModule.h	4156h	8-system triadic solver
UQFF8AstroTriadicModule.cpp	31681cpp	computeTriadicSolution()

5.2 Deduplication Status

- UQFFNebulaTriadicModule: appeared at L9597 (5-system) and L9990 (7-system/canonical) — L9990 version is canonical (larger, 320 cpp lines vs 289)
- UQFF8AstroSystemsModule: appeared at L10539, L10672, L11310, L11429, L11541 — L10539 version is canonical (383 cpp lines, largest)
- UQFFBuoyancyModule (7-system at L10102): 368-line cpp variant — skipped (UQFFBuoyancyModule.h/.cpp already exist from PAPER_479)

6. CP2 + CP4 Integration

Phase C (Session 126): CP2 Additions

Add three calculators to CondensedPhysics2.py: 1. UQFFBuoyancyAstroCalculator — Session 125 pending (CP2: 600→601) 2. UQFFBuoyancyCNBCalculator — Session 125 pending (CP2: 601→602) 3. IndividualSystemUQFF18Calculator — Session 126 (CP2: 602→603) 4. MultiSystemUQFFCompilerCalculator — Session 126 (CP2: 603→604) 5. HydrogenResonanceUQFFCalculator — Session 126 (CP2: 604→605)

Phase D (Session 125+126): CP4 Additions

- Session125GrokShare4e4d8be1f7HubCalculator (#104) — pending from Session 125
- Session126GrokShareBdfb3a05b06HubCalculator (#105) — this session

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